

TRAVELING SALESPERSON PROBLEM

Aleksey Sokolov Matr.: 12004091

Monte Carlo Methods
Supervisor: Sexty, Denes, Assoz. Prof. PhD
5. Februar 2025

Inhaltsverzeichnis

1	Introduction	3
2	fundamentals	3
2.1	Computational Complexity and Algorithmic Approaches	3
2.2	Physical and Industrial Applications	3
2.3	Theoretical Significance	4
3	Methods	4
3.1	Simulated Annealing	4
3.2	Genetic Algorithm	5

1 Introduction

The traveling salesman problem is a popular entry problem for computational optimization using probabilistic methods. It was first formulated in 1930 and is since then used as a benchmark for many optimization algorithms. Even though the problem is computationally difficult, many heuristics and exact algorithms are known, which can approximate the exact solution within a small fraction of 1 %. This document focuses on implementing two different approaches to finding the optimal path and then comparing their runtime and uncertainties their calculations. First we focus on implementing a simple Metropolis algorithm with simulated annealing, where we compare different annealing strategies with the use of temperature functions. The second part of this document a genetic algorithm is implemented and at last compared to the probabilistic approach.

2 fundamentals

2.1 Computational Complexity and Algorithmic Approaches

The factorial scaling of the TSP, given by $N!$ distinct routes, renders brute-force methods impractical beyond small system sizes. Exact algorithms, such as the *Held-Karp dynamic programming approach*, provide solutions but require exponential computational resources, limiting their applicability. As a result, most practical implementations rely on *heuristic and approximation algorithms*, which yield near-optimal solutions at significantly reduced computational cost.

Among the most effective heuristic methods are:

- **Greedy algorithms**, which construct solutions iteratively based on local distance minimization.
- **Metaheuristic approaches**, including *simulated annealing*, *genetic algorithms*, and *ant colony optimization*, which employ stochastic strategies to escape local minima.
- **Mathematical optimization techniques**, such as *integer linear programming relaxations* and *cutting-plane methods*, which refine feasible regions iteratively.

Although these methods do not guarantee exact solutions, they remain essential in real-world applications where computational constraints necessitate efficient approximations.

2.2 Physical and Industrial Applications

Optimization problems analogous to the TSP emerge in numerous domains, particularly in physics and engineering. Examples include:

- **Path optimization in transport networks**, where fuel efficiency and travel time minimization are critical.
- **Manufacturing processes**, where CNC machining and robotic assembly require minimizing tool movement to enhance efficiency.
- **Computational biology**, where the sequencing of DNA fragments or protein folding pathways presents combinatorial structures comparable to TSP-like models.

Beyond applied optimization, the TSP is frequently utilized as a *benchmark problem* for evaluating new algorithmic techniques, serving as a standard reference in computational complexity research.

2.3 Theoretical Significance

The TSP holds deep implications in computational complexity theory, particularly in relation to the *P vs NP problem*. As an NP-hard problem, the TSP is unlikely to have a polynomial-time solution unless $P = NP$, a hypothesis that remains one of the most significant open questions in theoretical computer science. A resolution to this problem would not only transform computational mathematics but also have profound consequences for cryptography, artificial intelligence, and algorithmic research.

3 Methods

There are three algorithms used in this project paper. In this section these three methods are explained and discussed.

3.1 Simulated Annealing

Simulated Annealing is a computational method that takes inspiration from solid state physics, namely from solid annealing and crystalization. As solids are heated they form different crystal structures. If left to cool local order emerges, driven by the tendency of a material to reduce internal stress and to attain a lower energy state. The same principle is applied to the a sequence of city positions where the order in which the cities are visited strongly influences the path length. In this approach, the length of sequentially ordered cities is considered the Free Energy of the system:

$$E(\{\mathbf{x}\}) = \sum_{i=1}^N |\mathbf{x}_{i+1} - \mathbf{x}_i| \quad \text{mit} \quad \mathbf{x}_{N+1} = \mathbf{x}_1. \quad (1)$$

In this, the $N!$ combinations of states (particular permutations) form a statistical ensemble, which corresponds to a partition function. The partition function encapsulates the statistical properties of the system in thermodynamic equilibrium. It serves as a function of thermodynamic state variables, such as temperature, and provides access to macroscopic quantities like free energy through its derivatives. The canonical partition function, commonly used in such contexts, describes a system in thermal equilibrium with its environment at a fixed temperature. Since direct computation of the partition function by summing over all possible states is infeasible due to the factorial growth of possible permutations, Simulated Annealing is employed to explore the configuration space.

$$Z = \sum_i e^{-\beta E_i} \quad (2)$$

One can use the Boltzman distribution to determine the probability of observing a state $\mathbf{x}$ at a given temperature and observed energy.

$$p_E(\{\mathbf{x}\}|T) = \frac{1}{Z} \exp \left\{ -\frac{1}{T} E(\{\mathbf{x}\}) \right\} \quad (3)$$

Since Z cannot be computed in a reasonable time for large systems, this distribution tells that a thermodynamic system favors lower energy states, especially at lower temperatures. This probability can be used to construct an acceptance probability for changing states as seen in eq. 4. By this definition negative Energy changes are always accepted, while positive are decreasingly unlikely with lower temperature. In combination with temperature variations this allows for efficient exploration of the phase space.

$$\min \left(1, \frac{p_E(x_2|T)}{p_E(x_1|T)} \right) \quad (4)$$

In order to traverse the energy landscape, a appropriate change must be proposed. This is realized by taking a random subpath of the cities sequence by generating two random indeces $(a, b) \in (1, N)$ and flipping it, as seen here:

$$K : S_1, \dots, S_{a-1}, \underbrace{S_a, S_{a+1}, \dots, S_{b-1}, S_b}_{old}, S_{b+1}, \dots, S_N \quad (5)$$

$$K' : S_1, \dots, S_{a-1}, \underbrace{S_b, S_{b-1}, \dots, S_{a+1}, S_a}_{new}, S_{b+1}, \dots, S_N \quad (6)$$

The advantage of this proposal is that one can efficiently calculate the change in energy without iterating over the whole array:

$$E(K') - E(K) = \overline{S_{a-1}S_b} + \overline{S_aS_{b+1}} - \overline{S_{a-1}S_a} - \overline{S_bS_{b+1}} \quad (7)$$

By applying the changes and comparing a uniform generated number with the acceptance probability one can minimize the energy. If one chooses a appropriate temperature function, which changes the temperature input every sweep (N^2), local minima can be explored. Even a global minima can be found, but this can not be verified even for relatively few cities.

3.2 Genetic Algorithm

This approach is derived from nature. It simulates natural evolution of a population of individual cities sequences. As in nature there are steps of inheritance ,mutation and natural selection. There are several ways to implement this concept and the used algorithm is a basic version with only the core steps. In this approach the cost function needs to be evaluated very often and the energy needs to be calculated in its entirety. At first a population from one path is created by shuffling the path N times and saving the individuals. This ensures genetic diversity at the start. The following steps are implemented, which are done each iteration:

- **Selection**
asdasd
- **Reproduction**
- **Mutation**

Literatur